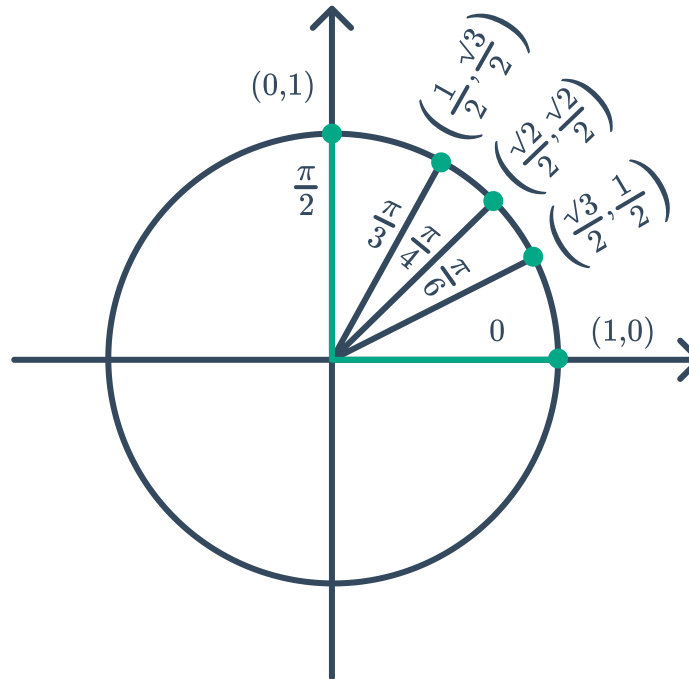


# Trigonometric functions on the unit circle



1 Consider the following diagram of the unit circle:



Find the exact value of the following:

a  $\cos \frac{\pi}{3}$

b  $\sin \frac{\pi}{3}$

c  $\sin \frac{\pi}{4}$

d  $\cos \frac{\pi}{6}$

e  $\cos \frac{\pi}{2}$

f  $\sin \frac{\pi}{2}$

g  $\cos 2\pi$

h  $\tan \frac{\pi}{6}$

2 Find the exact values of the following:

a  $\sin \left( \frac{7\pi}{6} \right)$

b  $\cos \left( \frac{2\pi}{3} \right)$

c  $\tan \left( \frac{7\pi}{4} \right)$

d  $\sin \left( \frac{19\pi}{6} \right)$

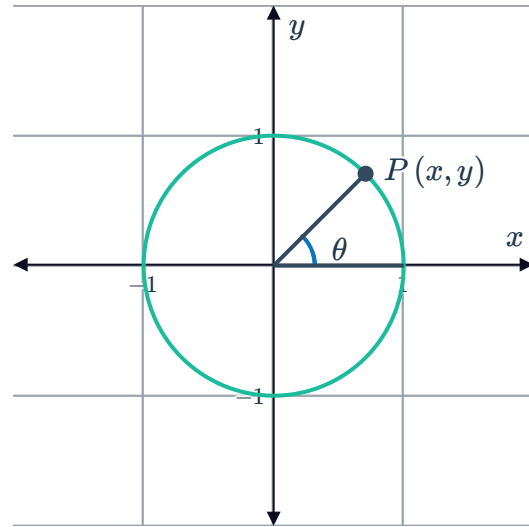
e  $\cos \left( -\frac{\pi}{6} \right)$

f  $\tan \left( \frac{11\pi}{4} \right)$

g  $\tan \left( -\frac{\pi}{6} \right)$

h  $\sin \left( -\frac{5\pi}{6} \right)$

- 3 The point  $P$  on the unit circle has the coordinates  $(x, y)$ .  
Write the following in terms of  $x$  and/or  $y$ :



- a  $\csc \theta$                       b  $\sec \theta$   
c  $\cot \theta$

- 4 Find the exact value of the following:

- a  $\sec(-30)^\circ$               b  $\cot 225^\circ$               c  $\csc 240^\circ$               d  $\sec 300^\circ$   
e  $\cot 585^\circ$               f  $\csc 945^\circ$               g  $\sec 945^\circ$               h  $\cot 945^\circ$

- 5 Find the exact value of the following:

- a  $\sec \frac{\pi}{6}$                       b  $\csc \frac{\pi}{3}$                       c  $\sec \left( \frac{2\pi}{3} \right)$               d  $\csc \left( \frac{4\pi}{3} \right)$   
e  $\cot \left( \frac{7\pi}{6} \right)$               f  $\sec \left( \frac{15\pi}{4} \right)$               g  $\csc \left( -\frac{2\pi}{3} \right)$               h  $\cot \left( \frac{25\pi}{6} \right)$

- 6 For each of the following functions, state:

- i The quadrants where the function is positive.  
ii The quadrants where the function is negative.

- a Sine function                      b Cosine function  
c Tangent function

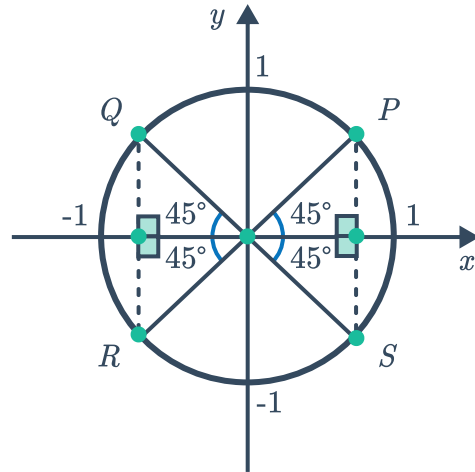
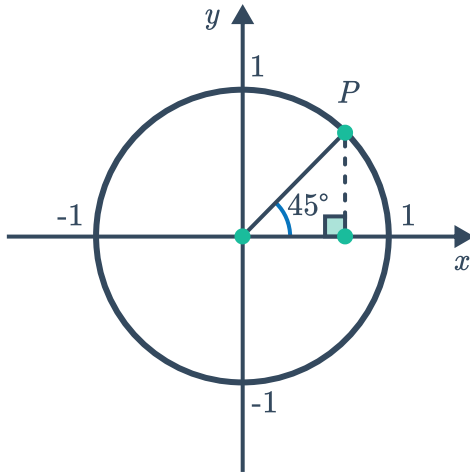
- 7 Each of the given points corresponds to the angle  $\theta$ , measured counterclockwise from the positive  $x$ -axis. For each point, find the following ratios:

- i  $\sin \theta$                       ii  $\cos \theta$                       iii  $\tan \theta$                       iv  $\csc \theta$   
v  $\sec \theta$                       vi  $\cot \theta$

- a  $P \left( \frac{1}{5}, \frac{\sqrt{3}}{5} \right)$               b  $P \left( -\frac{2}{5}, \frac{\sqrt{21}}{5} \right)$               c  $P \left( \frac{1}{\sqrt{5}}, \frac{-1}{\sqrt{5}} \right)$



- 8 The first diagram shows a unit circle with point  $P \left( \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$  marked on the circle. Point  $P$  represents a rotation of  $45^\circ$  counterclockwise around the origin from the positive  $x$ -axis:



- a Find the exact values of the following:
- i  $\sin 45^\circ$       ii  $\cos 45^\circ$       iii  $\tan 45^\circ$
- b On the second diagram, the coordinate axes shows a  $45^\circ$  angle that has also been marked in the second, third, and fourth quadrants. For each quadrant, find the measure of the relative angle:
- i Quadrant 2      ii Quadrant 3      iii Quadrant 4
- c The points  $Q$ ,  $R$  and  $S$  mark rotations of point  $P$  to the corresponding angles on the unit circle. State the exact coordinates of each point:
- i  $Q$       ii  $R$       iii  $S$
- d Write the following in terms of an equivalent ratio of  $45^\circ$ :
- i  $\sin 135^\circ$       ii  $\cos 225^\circ$       iii  $\tan 315^\circ$       iv  $\sin (-45)^\circ$
- e Now, find the exact value of the following:
- i  $\sin 135^\circ$       ii  $\cos 225^\circ$       iii  $\tan 315^\circ$       iv  $\sin (-45)^\circ$

- 9 In the given diagram,  $OA = 1$ , and the coordinates of  $A$  are  $(\cos \theta, \sin \theta)$ .  $\overline{CD}$  is tangent to the unit circle at  $A$ .  $\triangle AOB$  is similar to  $\triangle COA$  and  $\triangle ODA$ .

- Show that  $AC = \tan \theta$ .
- Given that  $\sec \theta = \frac{1}{\cos \theta}$ , show that  $OC = \sec \theta$ .
- Given that  $\csc \theta = \frac{1}{\sin \theta}$ , show that  $OD = \csc \theta$ .
- Given that  $\cot \theta = \frac{1}{\tan \theta}$ , show that  $AD = \cot \theta$ .

